WiFi Network Monitoring Mobile Application

Project Description

Many WiFi network owners do not have clear visibility into the devices connected to their network or the websites those devices access. Without proper monitoring tools, it becomes difficult to detect unknown devices, identify potential security risks, or understand network usage patterns.

This project aims to develop a **mobile-based WiFi network monitoring application** that allows users to monitor connected devices, analyze network activity, and track device usage in real time. The system collects network traffic information and displays it through a user-friendly mobile interface.

Objectives

- Detects devices connected to the WiFi network.
- Display usage information for each device including:
    - Packet usage
    - Download activity
    - Upload activity
    - Active sessions
- Provide historical records of devices that have connected to the WiFi network.
- Display WiFi network details such as:
    - Network download speed
    - Network upload speed
    - Internet Service Provider (ISP)
    - Network latency (Ping)

Target Users

- Home WiFi users who want to monitor devices connected to their network.
- Small office administrators who need basic network monitoring capabilities.

Overview

The WiFi Network Monitoring Mobile Application provides a simple way for users to monitor network activity through their mobile devices. The system detects connected devices and collects network usage information such as packets, downloads, uploads, and sessions.

The backend system processes network data and stores it in a database, while the mobile application retrieves this information through APIs and displays it in a user-friendly format.

Users can view:

- Real-time network activity
- Device usage statistics
- Historical device information

This application helps users improve network visibility, identify unknown devices, and understand how their network resources are being used.

System Architecture

System Workflow

1. Devices connect to the WiFi network.
2. The packet sniffer captures network packets from the network interface.
3. DNS requests and packet information are extracted from captured packets.
4. Device information such as MAC address, IP address, packet usage, uploads, downloads, and sessions is processed.
5. The backend server stores the processed data in the database.
6. The mobile application retrieves the data through API endpoints.
7. The application displays the information using graphs and device activity lists.

System Components

1. Packet Sniffer

Captures network traffic from the WiFi interface and extracts useful information such as DNS queries and packet statistics.

2. Backend Server

Processes captured packet data, manages device information, and provides API endpoints for the mobile application.

3. Database

Stores network activity data including:

- Device MAC address
- Device IP address
- Website domains visited
- Packet usage
- Upload and download statistics
- Device connection history

4. Mobile Application

Provides the user interface where users can:

- View connected devices
- Monitor network usage
- View device history
- Analyze network statistics using graphs

Technologies Used

Programming Languages

- Python
- JavaScript
- SQL

Backend

- FastAPI (Backend API)
- Scapy
- Uvicorn
- SQLALchemy

Frontend

- React Native
- React Navigation
- Lucide React
- Axios

Security

- JWT(python-jose)
- Password Hashing(passlib with bcrypt)

Database

- SQLite

Utilities

- psutil(system info)
- ping3(latency)
- Python-dotenv

Dev Ops/Tools

- Node.js (used to run React Native and manage frontend dependencies)
- Android Studio
- Postman
- Git

Installation Instructions

Requirements

Before running the system install:

- Python 3.10 or higher
- Node.js
- Android Studio
- USB Debugged Android Device or Android Emulator
- Git

(SQLite does not require installation because it is included with Python.)

Backend Installation

- Clone the repository
  - git clone https://github.com/MohomedImshan/Network_Project_BackEnd.git
- Navigate to the backend folder
  - cd Network_Project_BackEnd
- Create a virtual environment
  - python -m venv venv
- Activate the environment
  - venv\Scripts\activate
- Install dependencies
  - pip install -r requirements.txt
- Run the backend server
  - uvicorn main:app --reload

Mobile Application Installation (Frontend)

- Clone the repository
  - git clone https://github.com/MohomedImshan/Network_Project_App_Frontend.git
- Navigate to the folder
  - cd Network_Project_App_Frontend
- Install dependencies
  - npm install
- Start Metro server
  - npx react-native start --reset-cache
- Run Android application
  - npx react-native run-android
- The application will build and install on the connected Android emulator or device.

APK Build (Optional)

- To generate an APK file:
  - cd android
- Run
  - .\gradlew assembleRelease
- APK location:
  - android/app/build/outputs/apk/debug/app-debug.apk
- You can share this APK file to install the application on other Android devices.

Usage Instructions

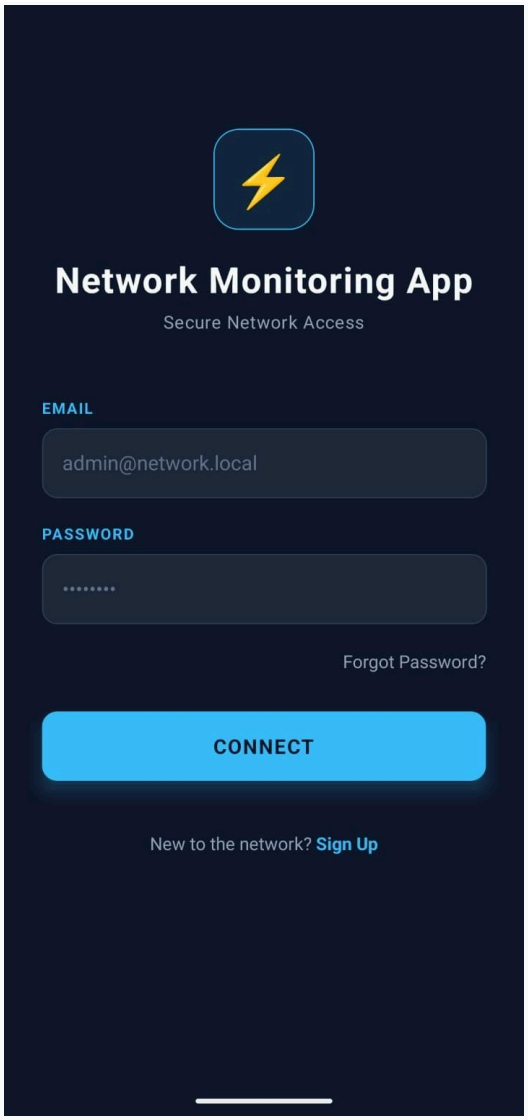
1. Start the backend server.
2. Start the mobile application.
3. Ensure both the backend server and mobile device are connected to the **same WiFi network**.
4. Ensure the env has the correct ip address.
5. Open the mobile application.

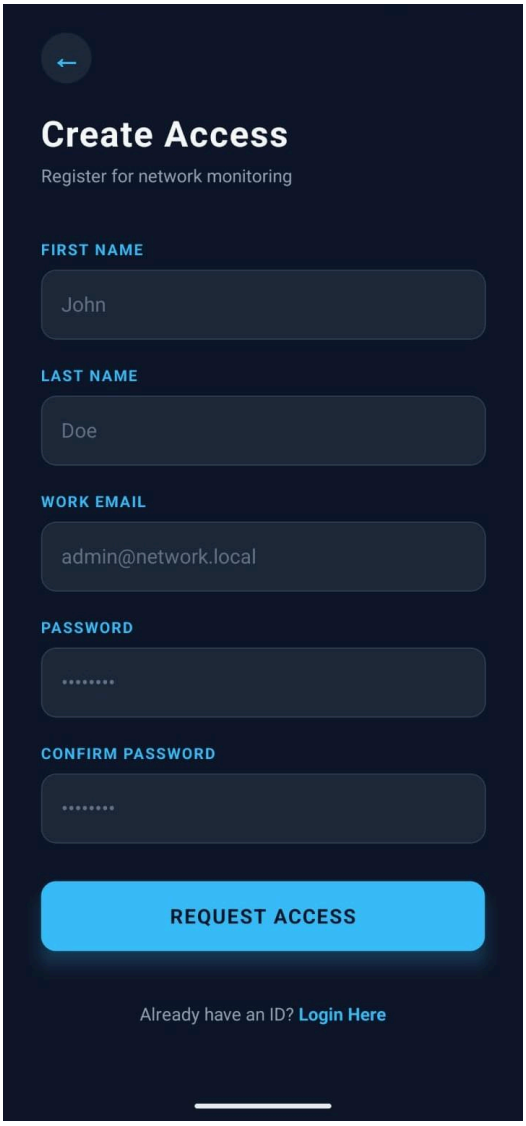
The application will display:

- Connected devices
- Device usage statistics
- Device visit history
- WiFi network information

Screenshots

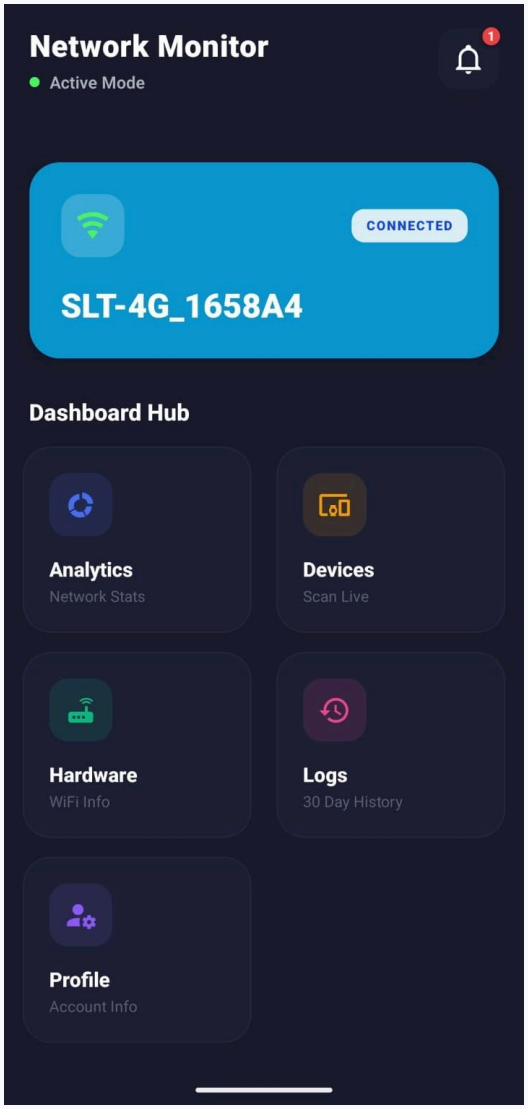
Log In and Sign Up:

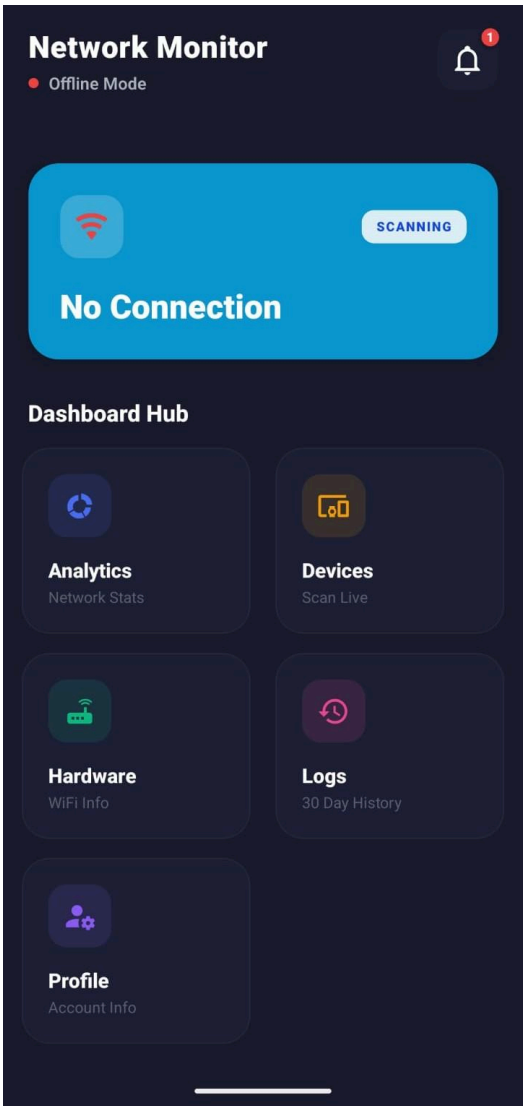




Home View (When wifi is connected/When Offline):

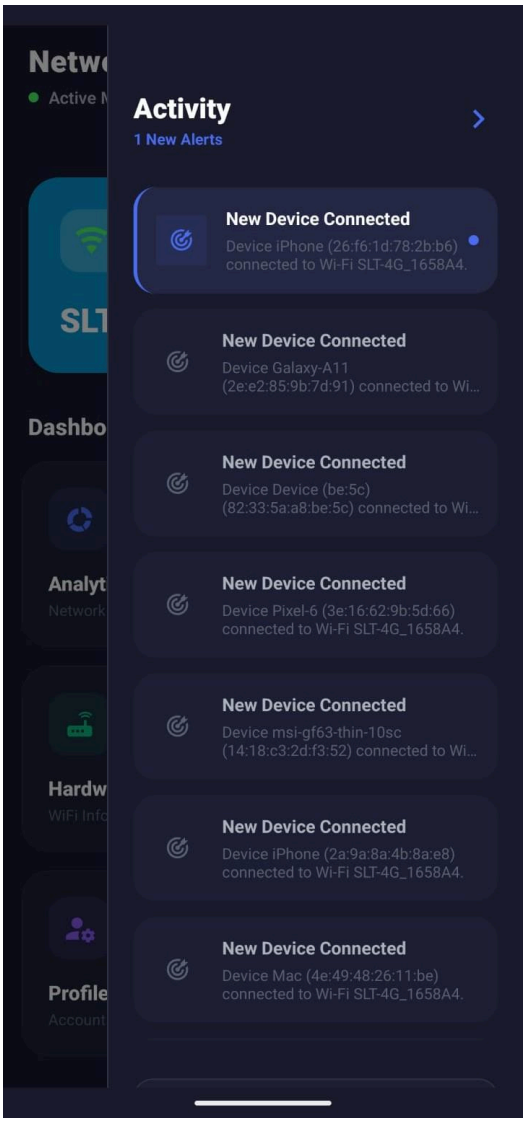
This is the front view of the application.When device connected to a wifi it shows the details.

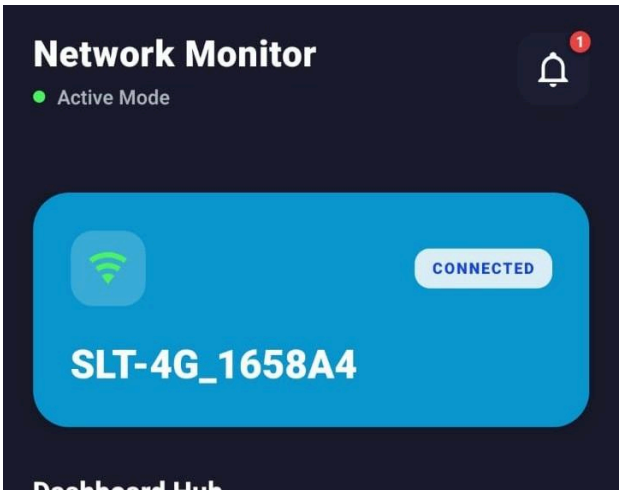




Notifications:

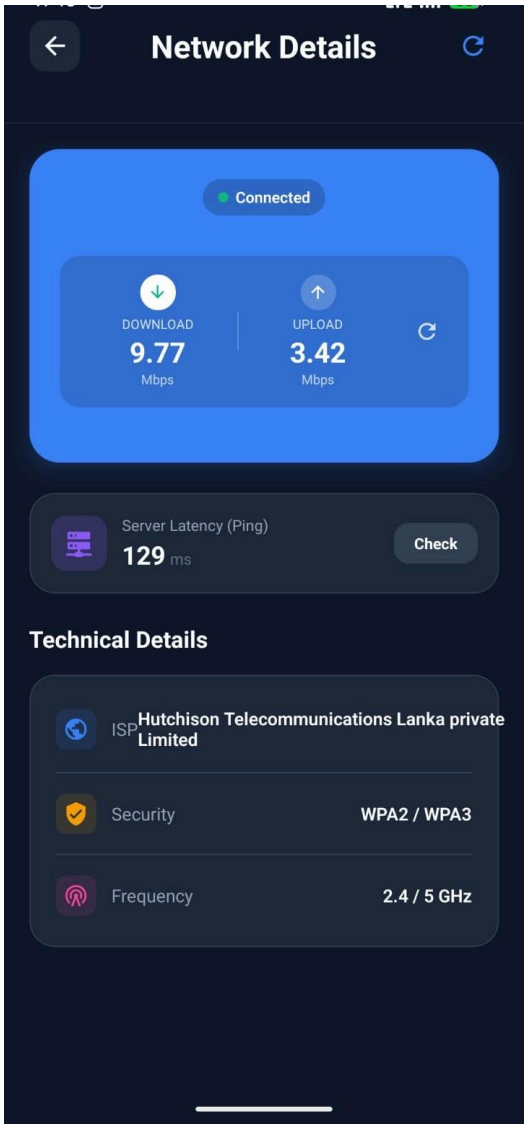
If a new device connects to the wifi this will notify the user and if there are malicious devices it will also notify.

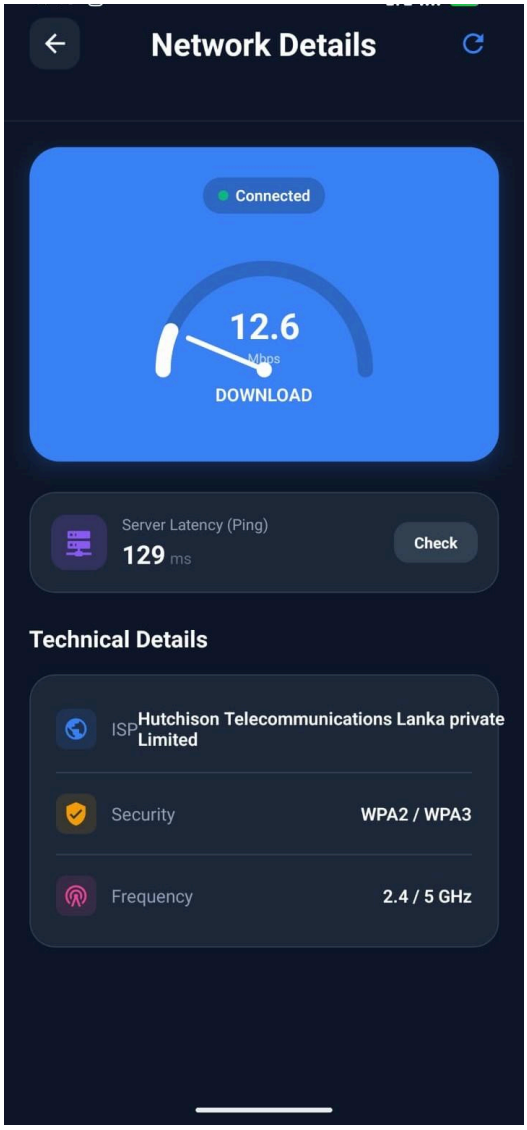




Network Stats:

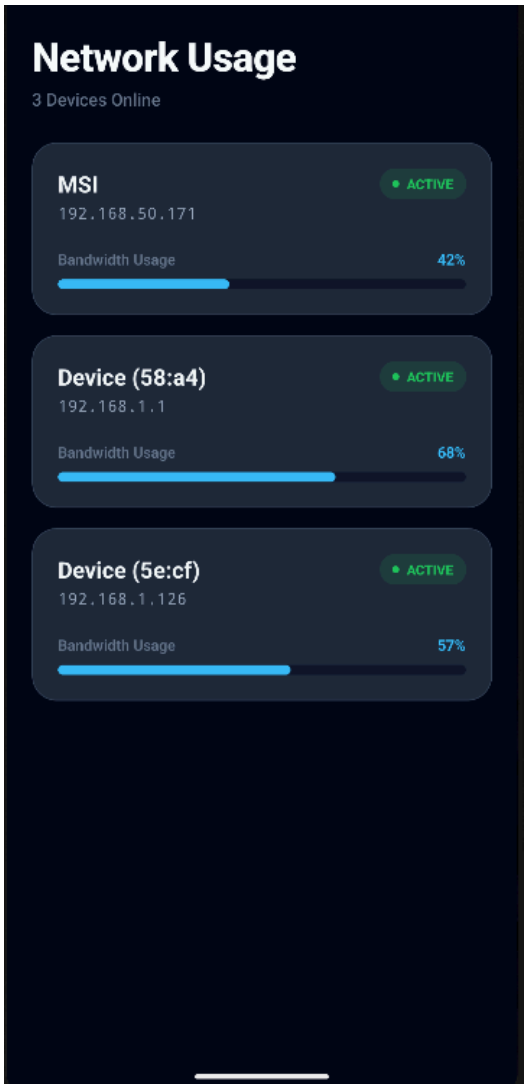
Shows the latency download upload speeds of the WIFI.





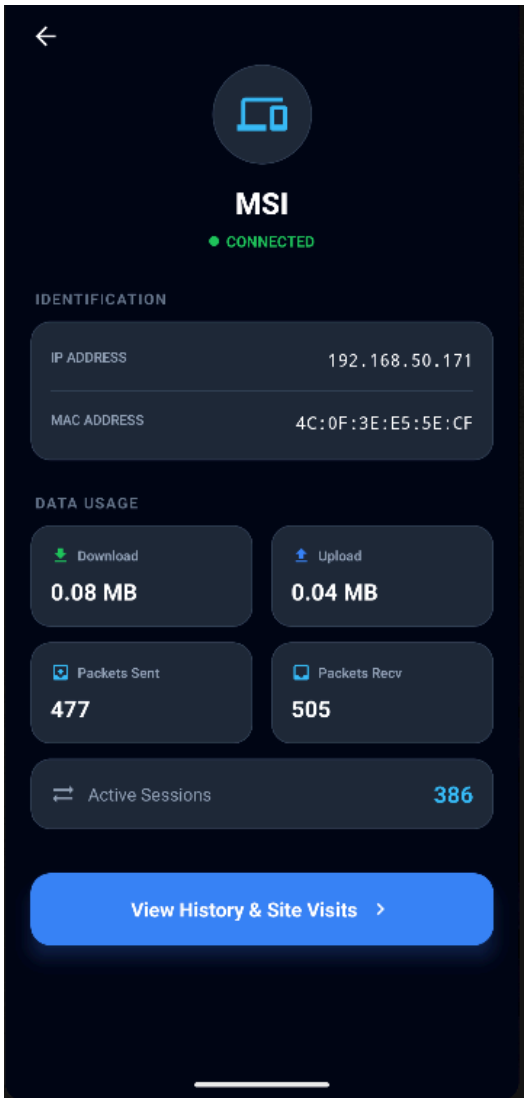
Active Devices:

Shows the connected devices to the wifi and usage of bandwidth.



Device Details and Usage:

Details of a particular device ,usage of packets ,downloads ,uploads,active sessions.



Site visited :

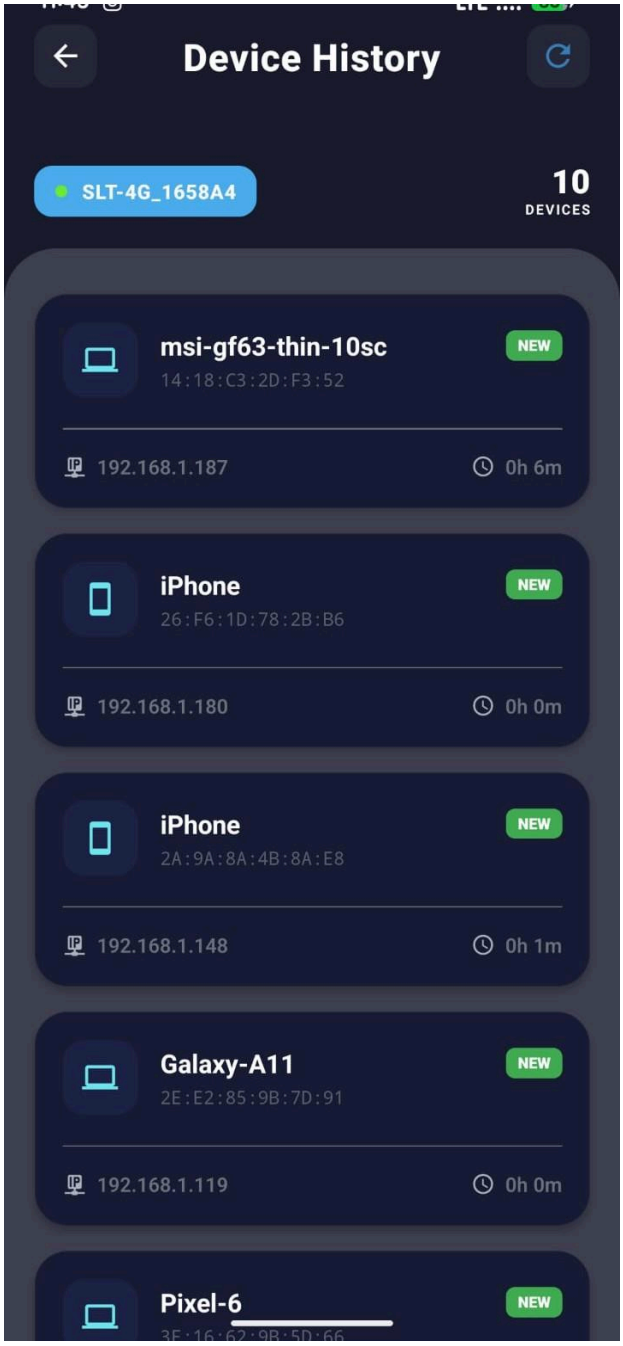
The site user has visited using the wifi.(Not accurate)

WIFI Details:

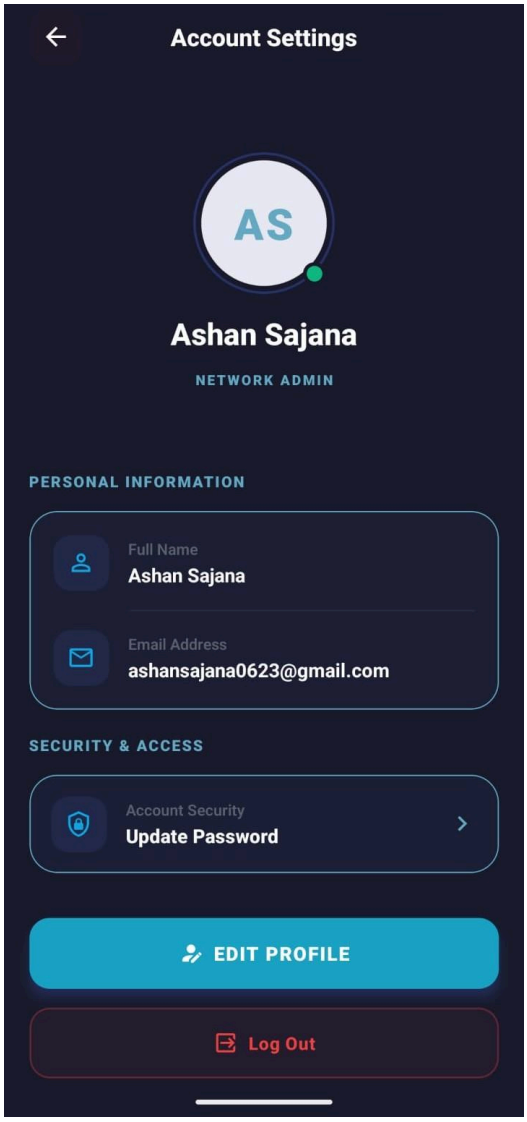
Details of the Internet service provider and the wifi.

Device History:

Connected devices history.



Account Details (Log out and Change Password):



Contributors

M.I.M Imshan

Team Lead
2021/CSC/042
Email: mohomedamccimshan@gmail.com

Responsibilities:

- Backend development
- System integration
- Code maintenance

W.G.P.B Wilagama

2021/CSC/042
Email: pasidubhagya20@gmail.com

Responsibilities:

- Backend development

R.P.A Sajana

2021/CSC/094
Email: ashansajana0623@gmail.com

Responsibilities:

- Frontend development

K.G.K Imalsha

2021/CSC/042
Email: kavishkaimalsha@gmail.com

Responsibilities:

- Authentication implementation
- Security features

License

This project is licensed under the **MIT License**.